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Applied Surface Science

Date: 1 September 2013

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Observation of spatter formation mechanisms in high-power fiber laser welding of thick plate

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10.1016/j.apsusc.2013.05.081 ↗

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Highlights

- The gauffers and shelf move down into the keyhole depth with direct observation.
- Raise of the constricted areas leads to melt accumulation around a narrow keyhole inlet with spatter ejection by vapor plume.
- Some spatters are accelerated through the directed vapor plume outside the keyhole.
- The downward flow, vapor burst with an open keyhole and gravity drive the melt off.
- The spatters are generated by the viscous friction drag through the keyhole exit.

Abstract

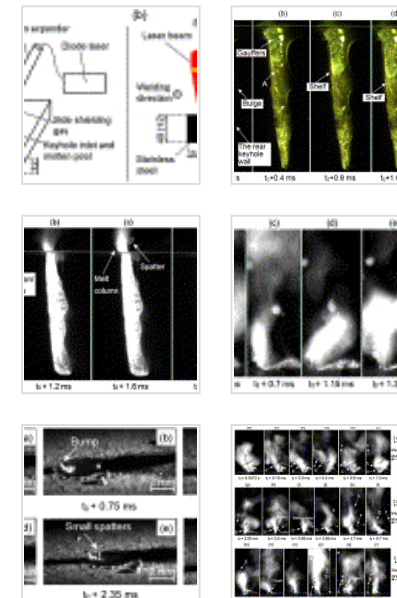
This paper aims to present the dynamic behaviors of spatter formation, and to clarify the spatter formation mechanisms in the high-power fiber laser welding of a thick plate at low welding speeds. We used a modified “sandwich” specimen to directly observe the geometry of the longitudinal keyhole wall. The dynamic behaviors of the keyhole, vapor plume, and melt pool with the formation of spatters were observed using high-speed imaging. The mechanisms of the formation of the spatter ejected from the top and bottom surfaces were analyzed. The recoil momentum associated with the energized vapor plume jet acts on the tips of the gauffers on the front keyhole wall and micro-droplets inside the keyhole, thereby resulting in the formation of high-speed micro-spatter. At partial penetration, the spatter ejected from the keyhole inlet is influenced mainly by the upward melt flow above the keyhole, melt displacement around the keyhole, and the strong shear stream of the directed vapor plume force. Moreover, some spatter droplets are accelerated through the vapor plume outside the keyhole. At full penetration of the melt, spatters are generated when the downward momentum of the melt due to downward flow and gravity, or vapor burst with an open keyhole, exceeds the surface tension forces. At full penetration of the keyhole, the crucial driving force for

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spatter generation is the viscous friction drag associated with high-speed motion of the energized vapor plume through the open keyhole. The welding process evolves into almost a cutting process at a lower welding speed.

 Video Video Video

Keywords

Fiber laser welding; Keyhole wall; Thick plate; Spatter; High speed imaging

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1. Introduction

When a laser beam with a high intensity ($>10^6 \text{Wcm}^{-2}$) impinges on a certain material, the material substrate surface starts to heat up. The surface temperature quickly reaches the melting point and a molten layer is formed. Subsequently, vaporization produces a recoil pressure that acts on the molten layer, removing molten material from the region ahead of the ablation front and consequently propagating the melt front into the metal bulk. Finally, a thin capillary (the so-called keyhole) is formed in the melt pool.

Spatter, which is the ejection of melt droplets from a weld pool, is a general defect observed in all welding processes. Melt pool ejections in the form of spatter will result in an unsteady appearance of the weld seam. Irregular weld surface features such as under-fill, under-cut, craters, and blowouts [1], [2] act as stress raisers that can severely reduce the mechanical properties of a weld.

With recent advances in the development of high-power fiber lasers with high focusability, it is expected that high-speed welding of thick plates can be realized. However, welding spatter is prone to be generated in high-power fiber laser welding [3],

[4]. As a consequence, a better understanding of the spatter formation mechanisms in high-power fiber laser welding is necessary to improve the weld seam quality.

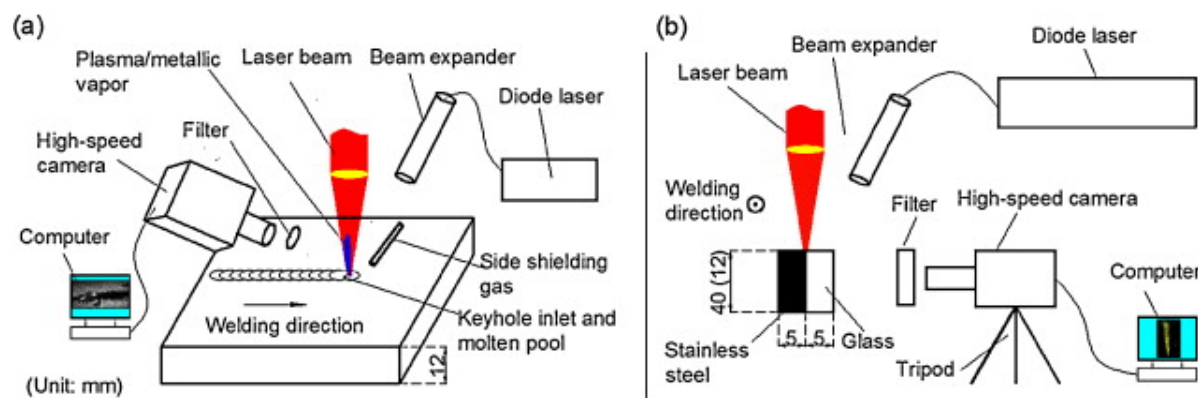
Previously, a few studies have investigated the spatter behavior in laser welding. Kawahito [4] observed the melt pool behavior and spattering in an under-filled weld bead produced by a 10kW fiber laser at a welding speed of 6mmmin^{-1} with a high-speed camera. Kawahito and his co-workers found that the generation of spatter was influenced mainly by a strong shear force of the laser-induced plume. Weberpals [5] observed the keyhole inlet coaxially and the melt pool around the keyhole using an obliquely angled high-speed camera. The welding speed ranged from 3mmmin^{-1} to 11mmmin^{-1} . Weberpals et al. found that the spatter was driven by increase in the inclination of the front keyhole wall and the consequent friction effect owing to the aerodynamic drag forces of the expanding metal vapor resulting from the local ablation inside the keyhole. Kaplan [6] provided a systematic description of different types of spatter phenomena occurring during laser welding, and also proposed a categorization system to facilitate the comparison and combination of research findings on spatter. Kaplan suggested the fundamental sequence of the spatter generation phenomenon as the following: local boiling--melt acceleration--redirection of fluid flow – accumulation of vertical momentum – droplet ejection. The experiment carried out by Kamimuki [7] indicates that the side gas jet contributes to the enlargement of the keyhole inlet resulting in stabilization of the metallic vapor escaping the keyhole, which greatly reduces the weld spatters. Semak [8] reported that the enhanced vaporization-induced recoil pressure induces the ejection of the melt from the interaction zone to form the spatter. In the reported literatures, all the imaging information of spatters captured by high-speed camera has been restricted to the behavior of vapor plumes outside the keyhole, melt pool, and keyhole entrance on the top surface. In this light, the spattering phenomena in the high-power fiber laser welding of thick plates at low welding speeds are not fully understood. Moreover, spatter ejected from the bottom surface has not been reported.

Therefore, in this study, using high-speed photography, we performed direct observation of the longitudinal keyhole, vapor plume, and melt pool dynamics during a high-power

fiber laser welding process carried out on a modified “sandwich” specimen. Moreover, the bead-on-plate welding process was also observed using a high-speed camera to clarify the spatter formation mechanisms in the high-power fiber laser welding of a thick plate.

2. Experimental set-up and materials

The experimental set-ups used for these experiments are shown in Fig. 1. Experiments were carried out with a continuous wave (CW) fiber laser (IPG YLS-10000). The maximum laser power is 10kW and the beam parameter product (BPP) is 7.5mmrad with a processing fiber of diameter 200μm. The laser beam emerging from the optical fiber end is collimated using a lens of focal length 150mm, and subsequently, it is focused on the specimen surface via a lens of focal length 300mm.



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Fig. 1. Schematics of experimental set-ups: (a) bead-on-plate welding; (b) welding with modified sandwich sample.

The material used is Type 304 austenitic stainless steel with thicknesses of 12mm and 5mm for the bead-on-plate welding and the modified sandwich sample, respectively. The modified sandwich sample consists of one sheet of stainless steel and one sheet of GG17 glass both of which have a size of either 40mm×40mm×5mm or 40mm×12mm×5mm, which are subject to partial penetration welding and full penetration welding,

respectively, as shown in [Fig. 1\(b\)](#). GG17 glass does not explode when subjected to 10-kW high-power fiber laser welding because of its low thermal-expansion coefficient and excellent heat resistance [\[9\]](#).

A high-speed video camera was used to observe the dynamics of the keyhole, the melt pool, and the laser-induced vapor plume, with its frame rate ranging from 5000fps (frames per second) to 20,000fps. The high-speed camera was operated under two different configurations for observation purposes. On the one hand, the camera was positioned laterally to overlook the top surface of the melt pool or to inspect the bottom surface of the melt pool at an oblique angle, as depicted in [Fig. 1\(a\)](#). On the other hand, the camera was mounted horizontally to view the longitudinal keyhole, the surrounding melt pool, and the laser-induced vapor plume laterally from the side of GG17 glass in the case of the modified sandwich sample, as depicted in [Fig. 1\(b\)](#). In order to observe the melt pool, a diode laser ($\lambda=808\text{nm}$) with a maximum power of 30W was used to illuminate the welding zone. Meanwhile, a filter was applied to the camera lens for selective observation (reducing the effects of the bright vapor-plume emission and the tremendous contrast between the welding pool and the surrounding zone). In our experiments, a bandpass filter with a transmission band of $808\pm 3\text{nm}$ and a filter with a transmission band ranging from 350nm to 650nm were positioned in front of the camera lens to observe the melt pool and the vapor plume, respectively.

The welding parameters used during the experiments are listed in [Table 1](#); the variation in these parameters thus resulted in seven welded samples with different penetration regimes. Nitrogen was used as a side-shielding gas for bead-on-plate welding. It was provided via a nozzle aimed at the top of the weld pool. The gas flow rate of nitrogen was set to 20lmin^{-1} .

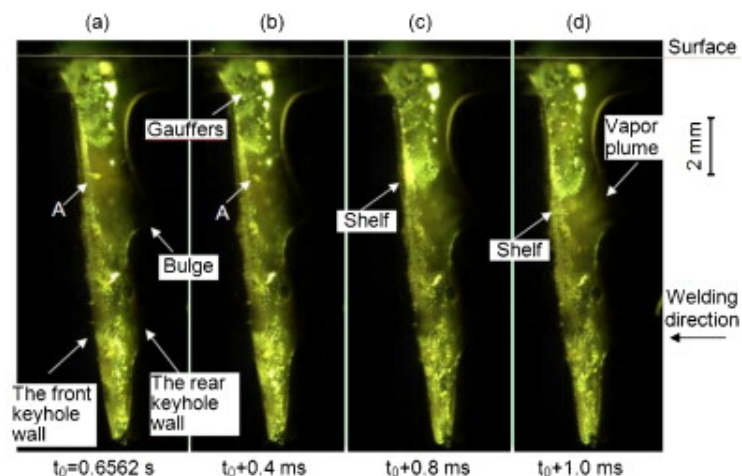
Table 1. Welding parameters with different penetration regimes.

Weld no.	Weld sample	Laser power (P) (W)	Weld speed (v) (mm min^{-1})	Defocus (Δ) (mm)	Penetration regime
1	Modified sandwich	10,000	1.2	5	Partial penetration of melt
2	Modified sandwich	10,000	1.5	0	Partial penetration of melt
3	Modified sandwich	10,000	0.9	-5	Full penetration of melt, sporadic penetration of keyhole
4	Bead-on-plate	10,000	3	-5	Partial penetration of melt
5	Bead-on-plate	10,000	3	-5	Partial penetration of melt
6	Bead-on-plate	10,000	0.9	-5	Full penetration of melt, sporadic penetration of keyhole
7	Bead-on-plate	10,000	0.75	-5	Full penetration of melt and keyhole

3. Results and discussion

3.1. Direct observation of keyhole, vapor plume, and melt pool dynamics with formation of spatter

Fig. 2 shows the image of the longitudinal keyhole during high-power fiber laser welding of the modified sandwich sample. Under the welding conditions mentioned in the figure caption, the diameter of the keyhole is ~ 1.32 mm at the entrance, and it subsequently decreases with the keyhole depth at partial penetration. The front keyhole wall is inclined slightly in the welding direction at an angle of $\sim 5.62^\circ$. As can be seen in Fig. 2 and Video 1, the keyhole wall is not smooth but fairly rough with the presence of typical “gauffer” structures formed owing to the surface tension forces [10], [11]. Moreover, the gauffers move down the keyhole owing to the recoil pressure of the vapor plume [10], [11], [13], [14], [15], [16]. One can also see micro-droplet being torn off the liquid wall of the keyhole (see Point A in Fig. 2(a)). The ejected micro-droplet flies toward the rear keyhole wall in a direction perpendicular to the front keyhole wall. This may be attributed to the small-scale capillary waves induced on the keyhole wall due to the capillary-evaporative instability of the liquid surface [12]. A bright zone is observed on the front keyhole wall, as shown in Fig. 2(c). This behavior may be caused by the moving liquid shelf, on which the laser beam irradiates directly, resulting in intense evaporation [11], [13], [17]. On the rear keyhole wall, we observe the formation of a bulge that subsequently moves up toward the surface. Therefore, we deduce that the vapor plume jet owing to the intense evaporation on the moving shelf at the front keyhole wall impinges on the rear keyhole wall to enhance the formation of the local bulge.

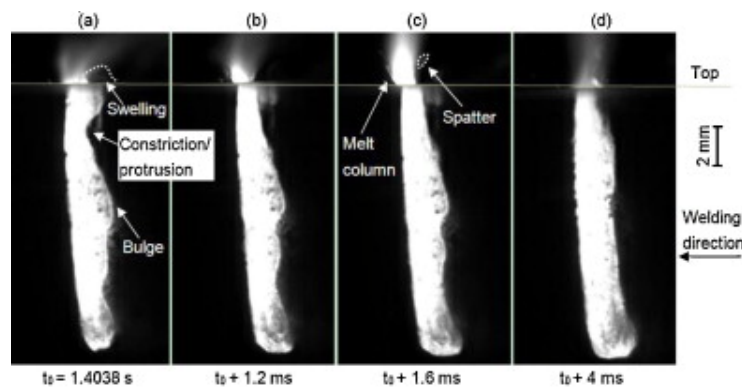


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Fig. 2. Direct observation of the longitudinal keyhole (laser power (P)=10kW, welding speed (v)=1.2mm min^{-1} , defocus (Δ)=5mm).

Fig. 3 shows simultaneously observed images of the longitudinal keyhole and the vapor plume from a side-view of the modified sandwich sample. The synchronized dynamics of the longitudinal keyhole and the vapor plume are captured to analyze the spatter formation. The front keyhole is almost vertical with a small inclining angle formed under the welding conditions described in the caption of Fig. 3. The rear keyhole wall shows a dynamic structural variation in its formation. A chain of bulges and constrictions is formed on the rear keyhole wall, as shown in Fig. 3(a)–(c) and Video 2. It is noteworthy that the formation of the constriction on the rear keyhole wall is also accompanied by the growth of a protrusion from the rear portion of the melt pool, which may be caused by the collision of the upward flow due to recoil pressure and the downward flow due to surface tension and hydrostatic pressure [18]. The constricted areas move upward along with upward flow of the molten metal. The upwelling melt accumulates to form a swelling behind the keyhole inlet, as shown in Fig. 3(a). Once the constricted area moves to the top surface, a strong vapor plume bursts through the narrow keyhole entrance, and this is accompanied by the splashing of the liquid metal from the keyhole, as shown in

Fig. 3(b) and (c). Simultaneously, the bulges and constrictions begin to diminish and disappear, as shown in Fig. 3(d). Thereafter, the keyhole is stabilized with the formation of a larger entrance and reduced vapor plume formation. We hypothesize that the fluctuation observed in the formation of the rear keyhole wall is mainly caused by the pressure imbalance at the rear keyhole wall due to the oscillations of the recoil pressure, surface tension, hydrostatic pressure, and fluid dynamic pressure. In addition, hydrodynamic instabilities such as the Rayleigh–Taylor instability [19], [20] and the Kelvin–Helmholtz instability [20], [21] also may be perturb the keyhole geometry. The irregular wave on the rear keyhole wall may be partially perturbed by the Rayleigh–Taylor instability.

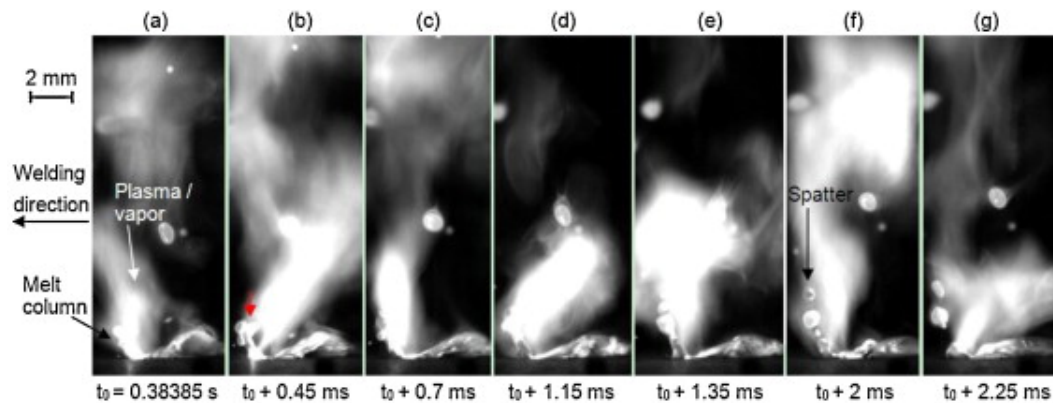


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Fig. 3. Images of the longitudinal keyhole and vapor plume with the formation of spatters ($P=10\text{kW}$, $v=1.5\text{mm min}^{-1}$, $\Delta=0$).

Fig. 4 shows images of the vapor plume and the surface melt pool with the ejection of spatters ahead of the keyhole. As can be observed, a melt column arises ahead of the keyhole, with a strong vapor plume ejected from the melt column along the column direction (Fig. 4(a)). The ejection of the vapor plume is perpendicular to the surface of the melt column in the direction opposite to the welding direction, as shown in Fig. 4(b). This may be attributed to the local evaporation occurring on the surface of the melt column. More interestingly, the tip of the melt column deviates in the same direction as the vapor

plume due to the shear force induced by the vapor plume, as indicated by the red arrow in Fig. 4(b). With the growth of the melt column ahead of the keyhole, the vapor plume is ejected from the keyhole along the column direction (Fig. 4(c) and (e)), and subsequently, the plume direction is perpendicular to the surface of the melt column in the direction opposite to the welding direction (Fig. 4(d)); the plume direction thus alternates in the abovementioned manner until the formation of the spatters ahead of the keyhole. Thereafter, the lower part of the rising melt column collapses back to the melt pool with a reduction in the vapor plume formation, as shown in Fig. 4(f) and (g).

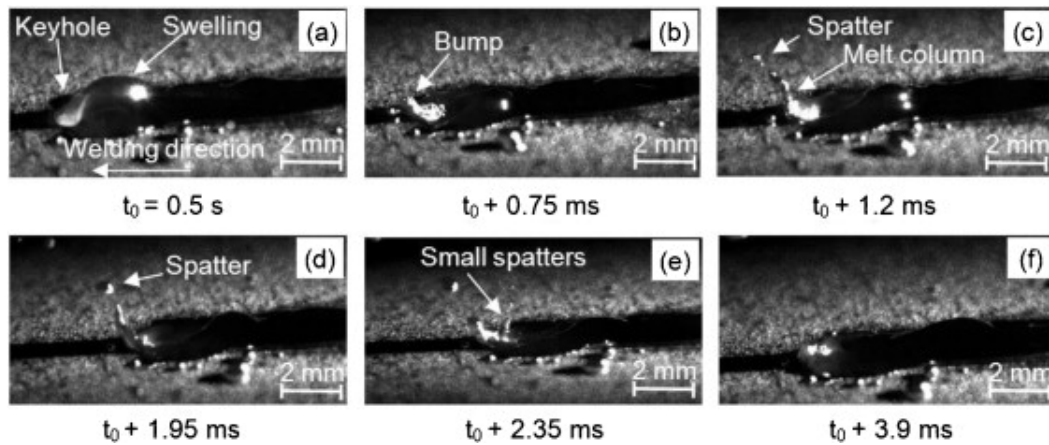


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Fig. 4. Images of the vapor plume and surface melt pool dynamics with the formation of spatters ahead of the keyhole ($P=10\text{ kW}$, $v=3\text{ mm min}^{-1}$, $\Delta=-5\text{ mm}$).

Fig. 5 shows images of the keyhole inlet and the surface melt pool under illumination by an auxiliary diode laser. After 0.5 s from the instant of laser irradiation, the keyhole inlet is stable and appears smooth and round. There is also a large swelling behind the keyhole, as shown in Fig. 5(a). Subsequently, the keyhole entrance is enlarged and wrinkled. A bump is generated around the keyhole entrance and the swelling is fairly small, as shown in Fig. 5(b). As the welding continues, the bright bump evolves into a tall melt column ahead of the keyhole, which is accompanied by the splashing of the upper part of the melt, as indicated in Fig. 5(c) and (d). Thereafter, the lower part of the melt is drawn back

into the melt pool, as shown in Fig. 5(e) and (f). As can be seen, occasionally, there are small spatters that are quickly ejected from the rear keyhole wall, as shown in Fig. 5(e).



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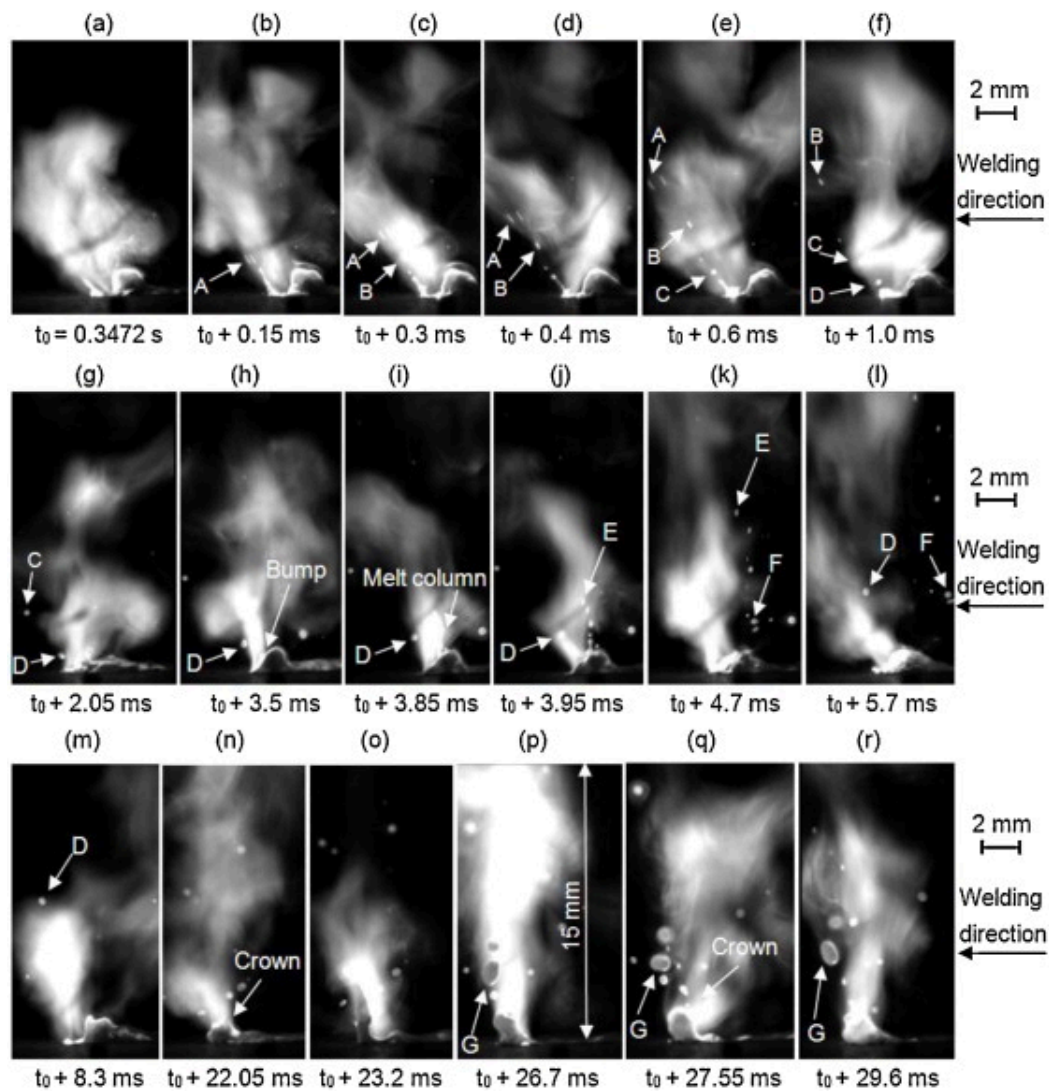
Fig. 5. Images of the keyhole inlet and surface melt pool dynamics with the formation of spatters ($P=10\text{ kW}$, $v=3\text{ mm min}^{-1}$, $\Delta=-5\text{ mm}$).

3.2. Formation mechanisms of spatters in high-power fiber laser welding of thick plate

3.2.1. Spatter generated around keyhole inlet at partial penetration

The high-speed observation images in Fig. 3, Fig. 4, Fig. 5 show that the spatters are ejected outward in various sizes at low welding speeds ($\leq 3\text{ mm min}^{-1}$). Fig. 6 shows images of the vapor plume and surface melt pool with the radial formation of spatters at the rim of the keyhole. Two droplets are directly ejected out of the melt pool with a high velocity of $\sim 11\text{ ms}^{-1}$, as indicated by Point A in Fig. 6(b). Along the forward direction of the emitted vapor plume (Fig. 6(a)–(c)), local evaporation may occur at the protrusion on the rear keyhole wall where the material absorbs the laser power, in a manner similar to that shown in Fig. 3(a). Further, the reason for the generation of the high-velocity micro-

spatter may be that the directed vapor plume jet impinges on the tips of the gauffers on the keyhole wall (see Fig. 2), or on micro-droplets inside the keyhole (see Fig. 2(b)).



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Fig. 6. Images of the vapor plume and surface melt pool dynamics with the formation of spatters around the keyhole ($P=10\text{ kW}$, $v=3\text{ mm min}^{-1}$, $\Delta=-5\text{ mm}$).

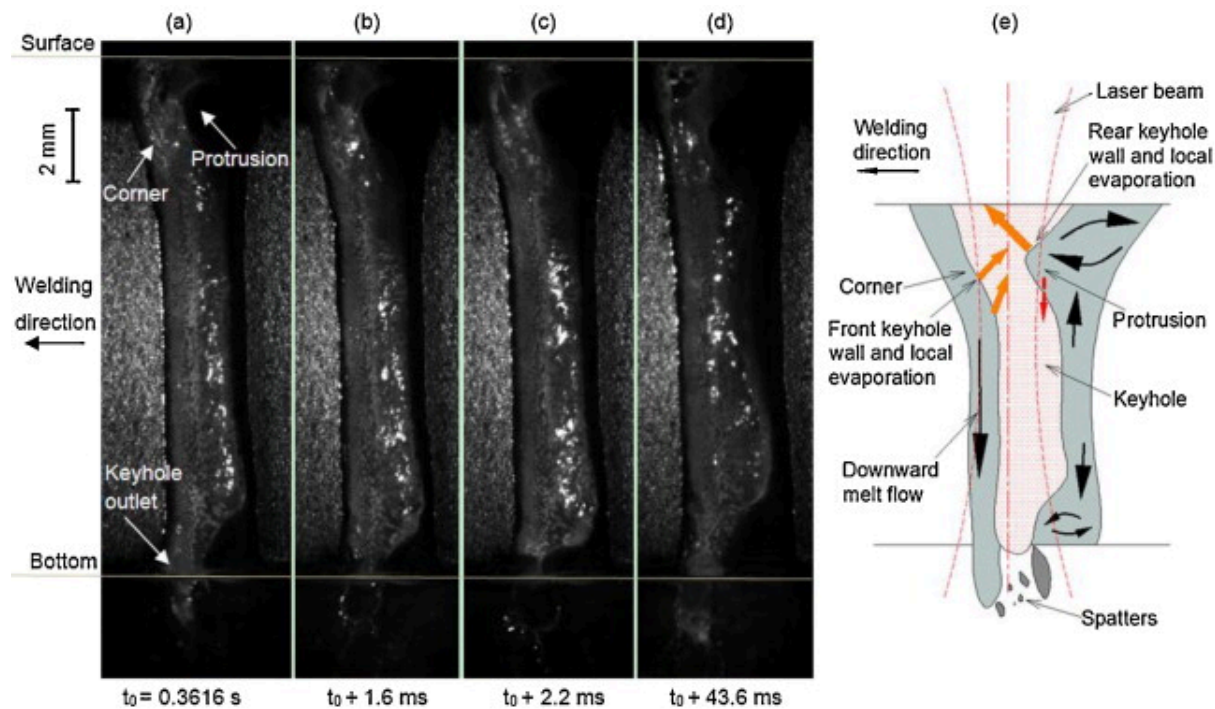
As can be seen in Fig. 6(a)–(f), with the growth of the melt column ahead of the keyhole, the tips of the melt column break away to continually form spatters with the ejection of a strong vapor plume clinging to the melt column. The moving velocity of the front spatter (Point B in Fig. 6(d)) is $\sim 6\text{ms}^{-1}$, and the following one (Point C in Fig. 6(e)) is $\sim 1.5\text{ms}^{-1}$. In particular, the last spatter (Point D in Fig. 6(f)) is ejected vertically from the melt pool with a velocity of $\sim 0.17\text{ms}^{-1}$. The spatter velocity increases up to $\sim 1.2\text{ms}^{-1}$ after it is forced through the directed vapor plume, as shown in Fig. 6(g)–(m). Obviously, there is an “acceleration process” of the spatter through the directed vapor plume outside the keyhole. An obvious melt displacement occurs around the keyhole entrance, which is accompanied by the splashing of a droplet and the disappearance of the swelling behind the keyhole, as seen from Fig. 6(g). As soon as a new swelling is formed behind the keyhole, a bump emerges from the rear keyhole wall and encounters the swelling with the formation of a strong vapor plume, as shown in Fig. 6(h). Thereafter, the bump grows along the vertical direction and forms a tall melt column with a vertical vapor plume, as shown in Fig. 6(i). As a consequence, the melt column is nearly simultaneously separated into several small droplets along different directions, as shown in Fig. 6(j). The escape velocities of the spatters vary with their location. The upper one (Point E in Fig. 6(j) and (k)) moves upwards with a velocity of $\sim 6\text{ms}^{-1}$; however, the lower one (Point F in Fig. 6(k) and (l)) moves rearwards with a velocity of $\sim 1.75\text{ms}^{-1}$. At the time instant of $t_0 + 22.05\text{ms}$, a substantial melt volume builds up, creating a melt “crown” around the keyhole (see Fig. 6(n)). A tall and large melt column arises from the side keyhole wall with a bulk mass of vapor plumes, as shown in Fig. 6(o). Subsequently, more intense vapor plume is expelled out of the keyhole with a height of more than 15 mm, which is accompanied with the detachment of several droplets from the melt column to form large spatters (see Fig. 6(p)). The large-sized spatters move with velocities ranging from $\sim 0.25\text{ms}^{-1}$ to $\sim 2.15\text{ms}^{-1}$. The size of the largest one (Point G in Fig. 6(p)–(r)) varies over time. The lower part of the melt column evolves into a new crown around the keyhole with the melt displacement, as shown in Fig. 6(q). As a result, several small spatters are ejected in all directions around the keyhole inlet with the formation of violent vapor plume, as shown in Fig. 6(q) and (r).

At low welding speed, there is a thick molten layer formed around the chimney-like keyhole owing to high thermal diffusion. The ablation recoil pressure in the keyhole exceeds the surface tension and hydrostatic pressure in the melt pool, thereby inducing an upward melt flow along the keyhole walls [18]. At the same time, the viscous drag associated with the high-speed motion of the energized vapor plume in the keyhole can also induce a strong upward melt motion in the direction parallel to the keyhole axis [22], [23], [24]. As a consequence, a substantial melt accumulates around the keyhole entrance to form a bump or crown, or even a tall melt column above the melt pool surface. Furthermore, the expanding stream of the directed vapor plume outside the keyhole could also exert a strong shear force on the melt column [4]. Eventually, when the local vertical momentum of the melt exceeds the surface tension force, the melt is ejected from the melt pool in the form of spatters. On the other hand, when the vertical momentum does not exceed the surface tension force, the melt is not ejected from the melt pool. Instead, the melt is eventually drawn back into the melt pool even though it rises above the pool.

3.2.2. Spatter ejected from bottom at full penetration

Fig. 7 illustrates the high-speed observation images of the longitudinal keyhole surrounded by molten metal with the ejection of spatters from the bottom during full-penetration welding of the modified sandwich sample. One can clearly observe from Fig. 7 that the upper part of the front keyhole is tilted slightly, while the lower part is almost vertical at a low welding speed. Interestingly, a corner is formed on the front keyhole wall near the top surface, which can be attributed to the formation of a protrusion on the upper part of the rear keyhole wall, as indicated in Fig. 3(a). In this case, the incident laser beam directly irradiates the protrusion, induces a local evaporation on the rear keyhole wall, and then results in the intense vapor plume jet impinging on the front keyhole wall, as shown in Fig. 7(e). As a consequence, the upper part of the front keyhole wall inclines forwards slightly with the formation of a corner on the front keyhole wall. In turn, the incident laser beam directly irradiates on the corner, which is followed by evaporation-

induced vapor plume jet that impinges on the rear keyhole wall and enlarges the keyhole entrance, as shown in Fig. 7(a)–(c) and (e).



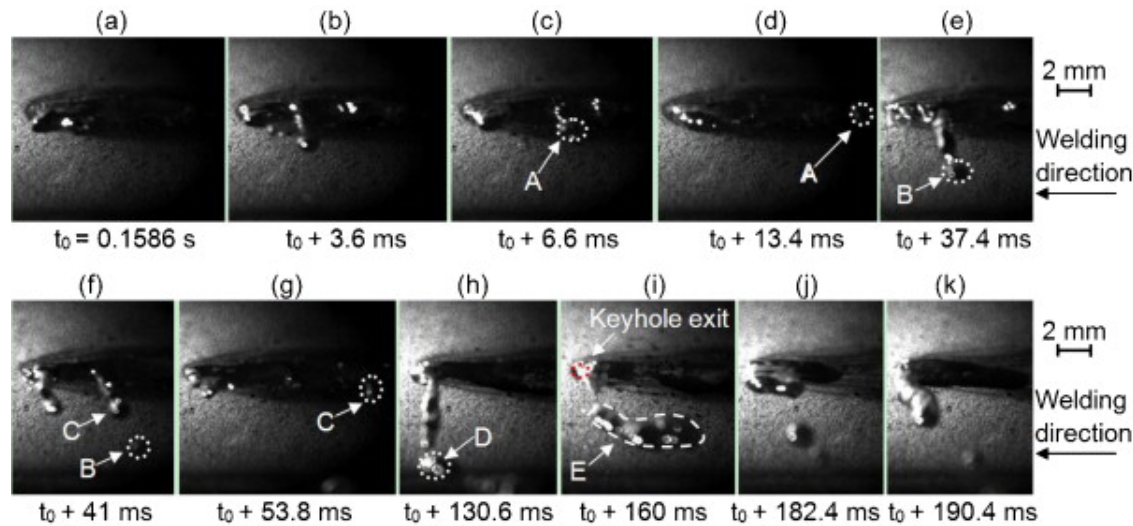
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Fig. 7. Images and schematic of the longitudinal keyhole surrounded by molten metal with the formation of bottom spatters ($P=10\text{ kW}$, $v=0.9\text{ mm min}^{-1}$, $\Delta=-5\text{ mm}$). (For interpretation of the references to color in text, the reader is referred to the web version of the article.)

On the other hand, the recoil pressure induced by the local evaporation of the upper surface of the corner or the moving shelf (Fig. 2(c) and (d)) drives the melt underneath downwards [14], [15], [16]. Moreover, the keyhole exit is narrowed by the surface tension force of the rear melt pool, which partially blocks the open keyhole. Consequently, a large amount of molten metal is blown out directly to form a saggy melt column at the bottom, as shown in Fig. 7(a). With a narrower keyhole exit, the melt is blown away in the form of

slender melt columns, as shown in Fig. 7(b) and (c). As the welding continues, the protrusion slides down, which is accompanied by the disappearance of the corner near the keyhole inlet, as shown in Fig. 7(d) and (e) (red dotted arrow).

Fig. 8 shows images of the bottom melt pool with the ejection of spatters being observed during full-penetration bead-on-plate welding from the bottom up. As can be seen in Fig. 8(a) and (b), the molten metal in the bottom melt pool is prone to be forced away in the direction opposite to the welding direction owing to the downward melt flow induced by the recoil pressure; it clings to the bottom melt pool under the action of surface tension. The saggy melt column moves backwards along the large melt pool, and later, it disintegrates into two parts. The upper part is pulled back into the melt pool due to the surface tension force; however, the lower part continues to move rearwards to form a spatter landing on the rear solidified weld surface (Point A in Fig. 8(c) and (d)). Therefore, we hypothesize that the molten metal below the bottom surface moves along a parabola traveling upwards with interactions among the downward flow induced by the recoil pressure and gravity, and the upward and rearward flow induced by surface tension with a large melt pool. The melt columns are apt to cling to the large melt pool, and continually disintegrate into spatters from the tip moving downwards (Point B in Fig. 8(e) and (f)) and rearwards (Point C in Fig. 8(f) and (g)). As the welding continues, the molten metal directly moves down to form a long melt column, which is accompanied by the formation of a small melt pool, and consequently the droplets separate from the melt column due to gravity (Fig. 8(h)) or vapor burst with an open keyhole (Fig. 8(i)). Once the small melt pool is formed, the upward flow of the melt due to surface tension cannot further flow rearwards (see Fig. 8(j)). As a consequence, a new melt column is formed, as can be seen from Fig. 8(k).

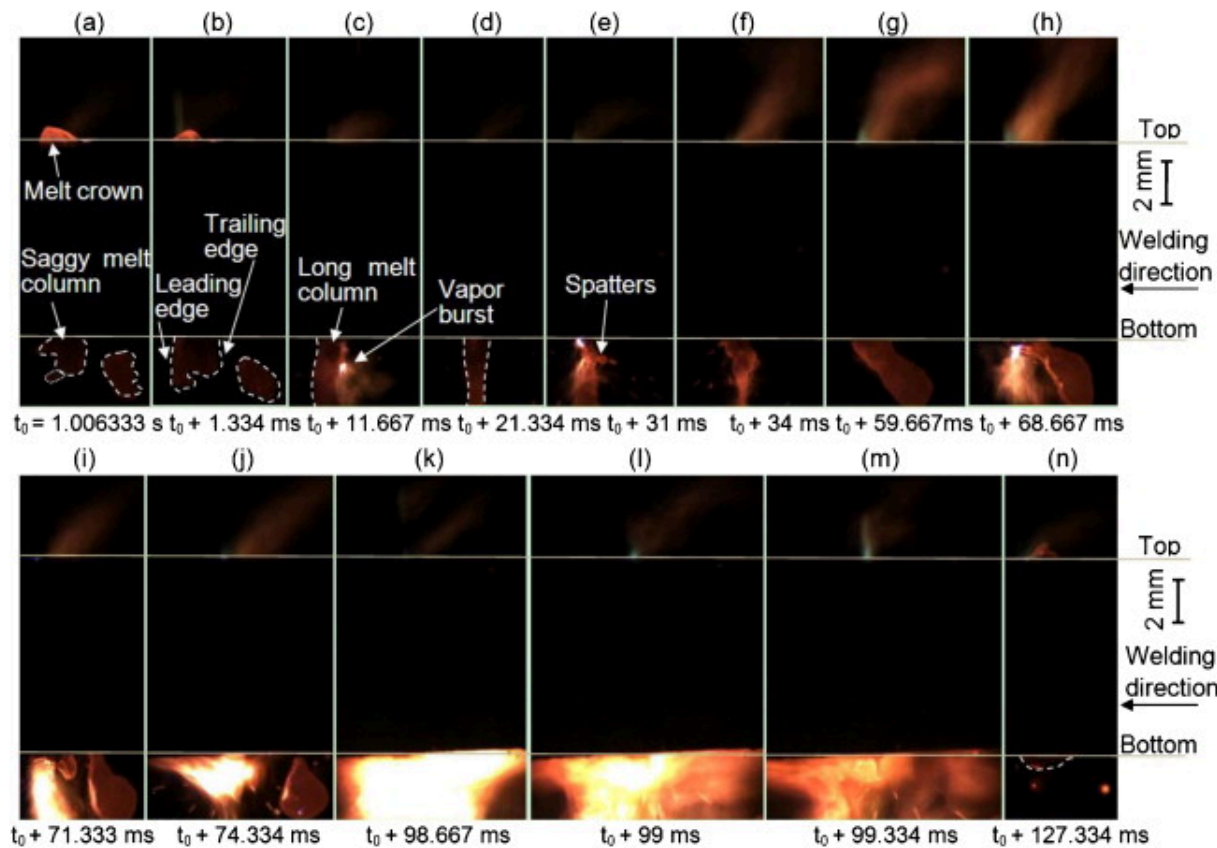


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Fig. 8. Images of the bottom melt pool dynamics with formation of bottom spatters viewed from bottom up ($P=10\text{kW}$, $v=0.9\text{mmmin}^{-1}$, $\Delta=-5\text{mm}$).

Fig. 9 and Video 4 show the high-speed observations of the surface melt pool and vapor plume dynamics with the ejection of spatters during full penetration bead-on-plate welding. A substantial melt flows downwards to cling to the bottom melt pool. This is accompanied by the build-up of a melt crown around the keyhole inlet, as shown in Fig. 9(a). The leading edge of the saggy melt column is longer than the trailing edge, as shown in Fig. 9(a) and (b). This indicates that the leading melt portion of the saggy melt column moves downwards owing to vigorous evaporation-induced recoil pressure, at a speed faster than the trailing one that moves upwards with the action of surface tension force. As can be seen from Fig. 9, the vapor plume is ejected from the keyhole mainly in the direction opposite to welding direction. Combined with the observation of the longitudinal keyhole with the modified sandwich sample in Fig. 7, we infer the existence of a corner or shelf on the front keyhole wall near the top surface in Fig. 9. Therefore, it is suggested that the evaporation-induced recoil force on the top surface of the corner or shelf drives the melt to flow downwards. Along with the accumulation of the saggy melt

column, the swelling melt around the keyhole inlet reduces in size and eventually disappears. At the same time, the saggy melt column with a certain volume (and mass) can cause the accumulation of more melt under the action of gravity. Consequently, the downward flow of the molten metal is robustly sustained, resulting in the formation of a long melt column hanging from the bottom melt pool, as shown in Fig. 9(c). It is worthwhile to point out that the long vertical melt column exists for a large interval of time, as observed from Fig. 9(c)–(e). Such behavior may be attributed to two main factors. On the one hand, once the absorbed energy inside the keyhole increased, the keyhole is apt to penetrate deeper into the vertical melt column resulting in an increase of the inner area of the keyhole. In other word, the action area of the vapor plume inside the keyhole will increase with the enhanced evaporation, leading to a stable pressure of the vapor plume inside the keyhole. On the anther hand, the pressurized vapor plume is constrained by the surrounded melt column occasionally, as indicated in Fig. 9(c), which is similar with the formation mechanism of porosity due to keyhole collapse [3].



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Fig. 9. Images of the surface melt pool and vapor plume with the ejection of spatters ($P=10\text{kW}$, $v=0.9\text{mmmin}^{-1}$, $\Delta=-5\text{mm}$).

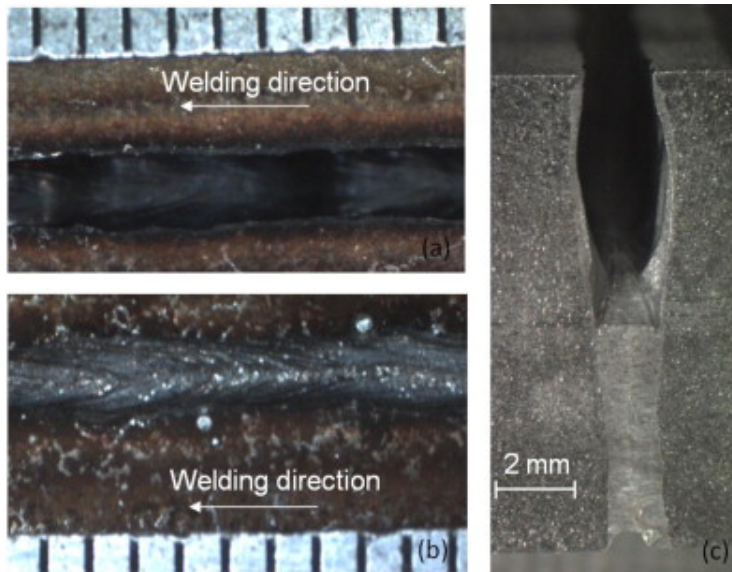
As the welding continues, the vapor plume inside the keyhole occasionally erupts from the melt column near the bottom surface; this is accompanied by the formation of many small spatters, as shown in Fig. 9(e). Thereafter, the flowing melt column undergoes necking and becomes slender, as shown in Fig. 9(f). Subsequently, the lower part of the long melt column breaks away from the melt pool at the neck due to the action of gravity. The remaining saggy melt grows continuously with the injection of downward melt flow from the internal melt pool, and it inclines in the direction opposite to the welding

direction, as indicated in Fig. 9(g). The vapor plume bursts out of the tilted melt column easily with a thin melt film at the bottom of the keyhole, as shown in Fig. 9(h). Subsequently, the large melt column moves rearwards due to the surface tension force (see Fig. 9(i)) and then narrows near the bottom surface of the substrate to form a big spatter, as shown in Fig. 9(j).

In order to maintain an open keyhole, the vapor pressure of evaporation should be greater than the closing pressure resulting from the local surface tension [24]. This fact is consistent with the observation of a substantial amount of energized vapor plume emerging from the bottom of the open keyhole; the amount of vapor ejected from the bottom is considerably more than that ejected out of the keyhole entrance from the top, as shown in Fig. 9(h)–(l). Further, a tremendous vapor plume escapes from the keyhole exit and expands outwards extensively below the bottom surface (see Fig. 9(k)) accompanied by several spatters quickly ejected away in all directions, as seen in Fig. 9(l) and (m). As in the case of the spatters ejected from the keyhole entrance, the viscous drag associated with high-speed motion of the energized vapor plume through the open keyhole can also induce a strong downward melt motion to generate spatters from the bottom. Our observations affirm that there is an immediate closing of the open keyhole without the ejection of spatter (see Fig. 9(n)). This may be attributed to the drop in pressure with the escape of the pressurized vapor, and the loss of the laser energy through the open keyhole [25]. These observations also indicate that the conditions for this welding process form a critical threshold for keyhole penetration.

As previously mentioned, the main driving forces causing the spatter generation at full penetration of the keyhole are the downward melt flow caused by evaporation-induced recoil pressure acting on the corner or the shelf on the front keyhole wall and the viscous friction drag associated with high-speed motion of the energized vapor plume through the open keyhole. Thus, it is reasonable to hypothesize that the downward flow along the front keyhole together with an open keyhole is sufficient to autogenously eject molten metal away from the bottom. Fig. 10 shows the geometries of the weld obtained at a high laser power and low welding speed. The upper part of the weld is not a joint but a narrow

kerf with a depth of ~ 6.42 mm. The welding process almost evolves into the cutting process. Last but not least, our experimental results also provide a good foundation for understanding the laser ablation cutting process [26].



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Fig. 10. Geometries of the weld obtained at a high laser power and a low welding speed: (a) top surface, (b) bottom surface, and (c) cross-section ($P=10\text{ kW}$, $v=0.75\text{ m min}^{-1}$, $\Delta=-5\text{ mm}$).

4. Conclusion

We studied the 10-kW high-power fiber laser welding of a thick plate at low welding speeds via high-speed imaging in order to explore the formation mechanisms of spatters. The following conclusions can be drawn:

The side keyhole wall and liquid motion along the keyhole wall are directly and clearly observed in the case of the modified sandwich specimen. The gauffers and shelf on the keyhole wall are observed moving down into the keyhole depth.

The formation of a bulge and constriction/protrusion on the rear keyhole wall is directly observed. The constricted areas move upward with the formation of a narrow keyhole entrance, thereby resulting in the splashing of the melt with a strong vapor plume burst.

At partial penetration, the recoil momentum associated with the energized vapor plume jet acts on the tips of the gauffers on the keyhole wall, and micro-droplets inside the keyhole torn off the keyhole wall, resulting in the formation of high-speed micro-spatters.

At partial penetration, the spatters ejected from the keyhole are attributed to the formation of a crown or melt column above the keyhole, melt displacement around the keyhole, and more importantly, a strong shear stream of directed vapor plume force blowing out of the keyhole inlet. Significantly, there is an “acceleration process” of the spatter through the directed vapor plume outside the keyhole.

At full penetration of the melt, the melt underneath the corner or the shelf is quickly driven down the front keyhole wall by the evaporation-induced recoil pressure, and this melt clings to the bottom melt pool. Once the downward momentum of the melt due to downward flow and gravity, or vapor burst with an open keyhole, exceeds the surface tension forces, spatters are generated.

At full penetration of the keyhole, the crucial driving force for spatter generation is the viscous friction drag associated with high-speed motion of the energized vapor plume through the open keyhole. The welding process evolves into almost a cutting process at a lower welding speed.

Acknowledgments

The authors are grateful to the financial support from the National Natural Science Foundation of China (No. [51175165](#)) and the Key National Science and Technology Project (No. [2013ZX04001131](#)).

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